



Task 0 - Setting the Baseline

Because of the upcoming holidays and the fact that you will probably not be able to work together, we have decided to make the first task all about getting comfortable with the code base of this project. This will include multiple things we recommend getting accustomed to, to be fully prepared for Task 1 in January. The prerequisites will be the following:

A: Learning CMD

This is the basic knowledge every coder requires. It is the ability to swiftly find your way and achieve certain goals within a command prompt! Therefore we urge you to get used to common commands within the terminal/powershell/command prompt. To get a quick overview, we recommend using an AI LLM. For more detailed information, please refer to official documentation released by Microsoft, Apple, ...

B: Installing Libraries

Having learned basic cmd knowledge we will now use the terminal to install necessary libraries. Please install the following software and make sure that the entire group is working with the same versions:

- GitHub
- OpenCV
- Python
- Microsoft VScode
- C++ together with the compiler of your choice
- CMake

C: Learning C++

Similar to large scale industry projects, project Eagle will be coded in C++. Although C++ is different from C, both base on very similar concepts which need to be understood. For this we recommend using the documents of the lectures as well as this YouTube Tutorial. Of course you can use any other material you would like. We also urge you to use an IDE such as Visual Studio or Visual Studio Code. Also make sure to get a thorough understanding of CMake. This is important to be able to run large project in C++.

D: Learning GitHub

Learning GitHub can be a pain ... always. However, when understood correctly it is one of the most useful tools you will come across when programming. This project keeps this in mind. For you to really get used to the GitHub workspace we have decided to base this whole project almost entirely inside GitHub, in this case a Project Eagle Repository. For this not to end in complete disaster it is very important to understand how GitHub works! There are numerous resources to learn about GitHub online. We recommend a combination of studying theory and learning by doing. Please be aware that you can lose ALL your progress with GitHub if you aren't careful!

E: Setting up a code base

For the last part of Task 0 we want you to create the basic code base for project Eagle. For this we want you to include necessary libraries into a main which for now should only print a message updating its status ("Running main ...", "Stopping main ...") After the GitHub meeting you will then also create your GitHub repository for the project. The detailed structure will also be explained in the meeting.